Final Project

2026-03-07

```r
prior_eqs <- read_csv("data/earthquakes_before_1950.csv")
```

```
## Rows: 39 Columns: 22
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (8): magType, net, id, place, type, status, locationSource, magSource
## dbl  (6): latitude, longitude, depth, mag, depthError, magError
## lgl  (6): nst, gap, dmin, rms, horizontalError, magNst
## dttm (2): time, updated
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

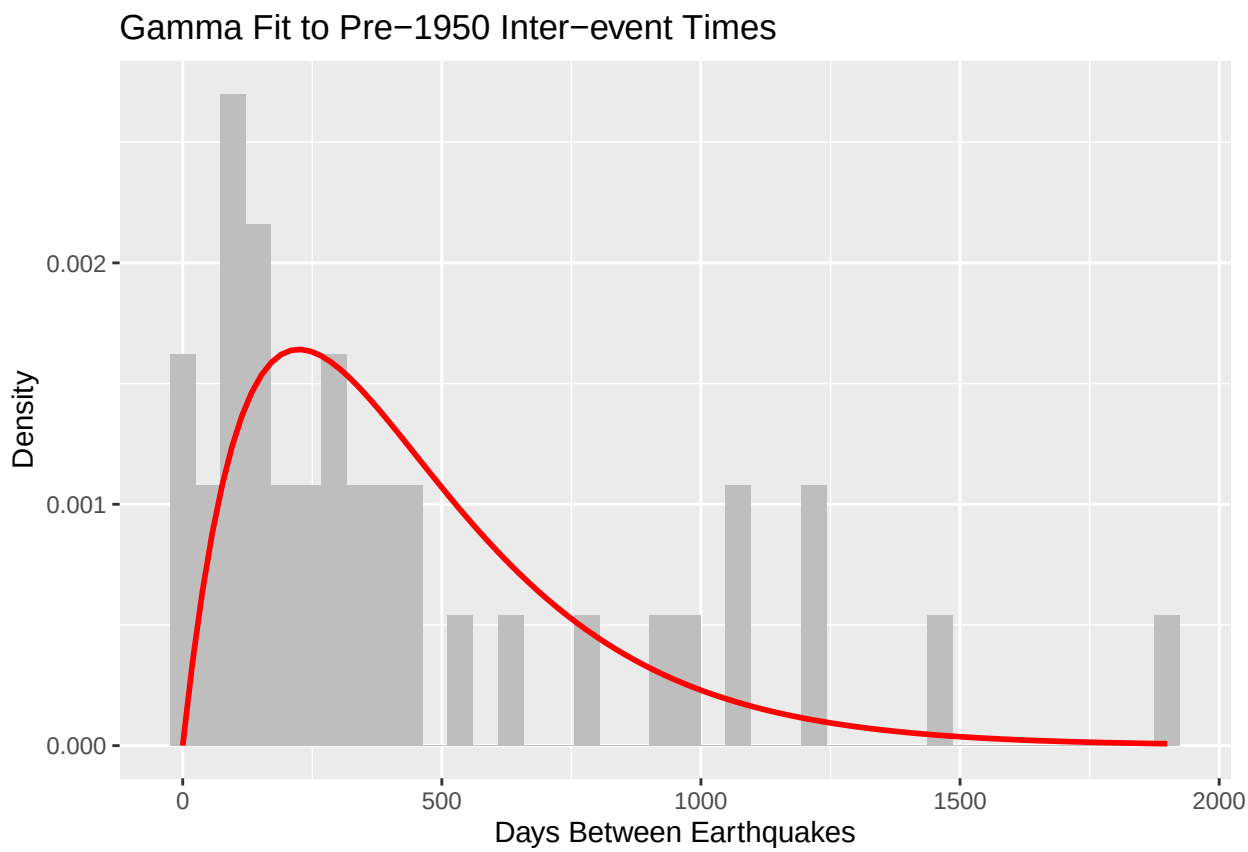
```r
prior <- prior_eqs[, c("time", "mag")]
prior$datetime <- as.POSIXct(prior$time, format = "%Y-%m-%dT%H:%M:%OSZ", tz = "UTC")
prior <- prior[order(prior$datetime), ]
prior$intertime <- c(NA, as.numeric(
  difftime(prior$datetime[-1],prior$datetime[-nrow(prior)], units="days")))
prior$Display <- round(prior$intertime, 2)
head(prior, 1)
```

```
## # A tibble: 1 x 5
##   time                  mag datetime            intertime Display
##   <dttm>              <dbl> <dttm>                 <dbl>   <dbl>
## 1 1903-01-04 05:07:00     8 1903-01-04 05:07:00        NA      NA
```

```r
#T ~ Gamma(2, theta)
between_time <- prior$intertime[!is.na(prior$intertime)]

#E(T)
E_T <- mean(between_time)

shape = 2
#E(T) = shape/rate
rate <- 2/E_T


ggplot(data.frame(T = between_time), aes(x = T)) +
  geom_histogram(aes(y = ..density..), bins = 40, fill = "grey") +
  stat_function(fun = dgamma,
                args = list(shape = shape, rate = rate),
                color = "red",
                linewidth = 1) +
```

```r
  labs(
    title = "Gamma Fit to Pre-1950 Inter-event Times",
    x = "Days Between Earthquakes",
    y = "Density"
  )
```

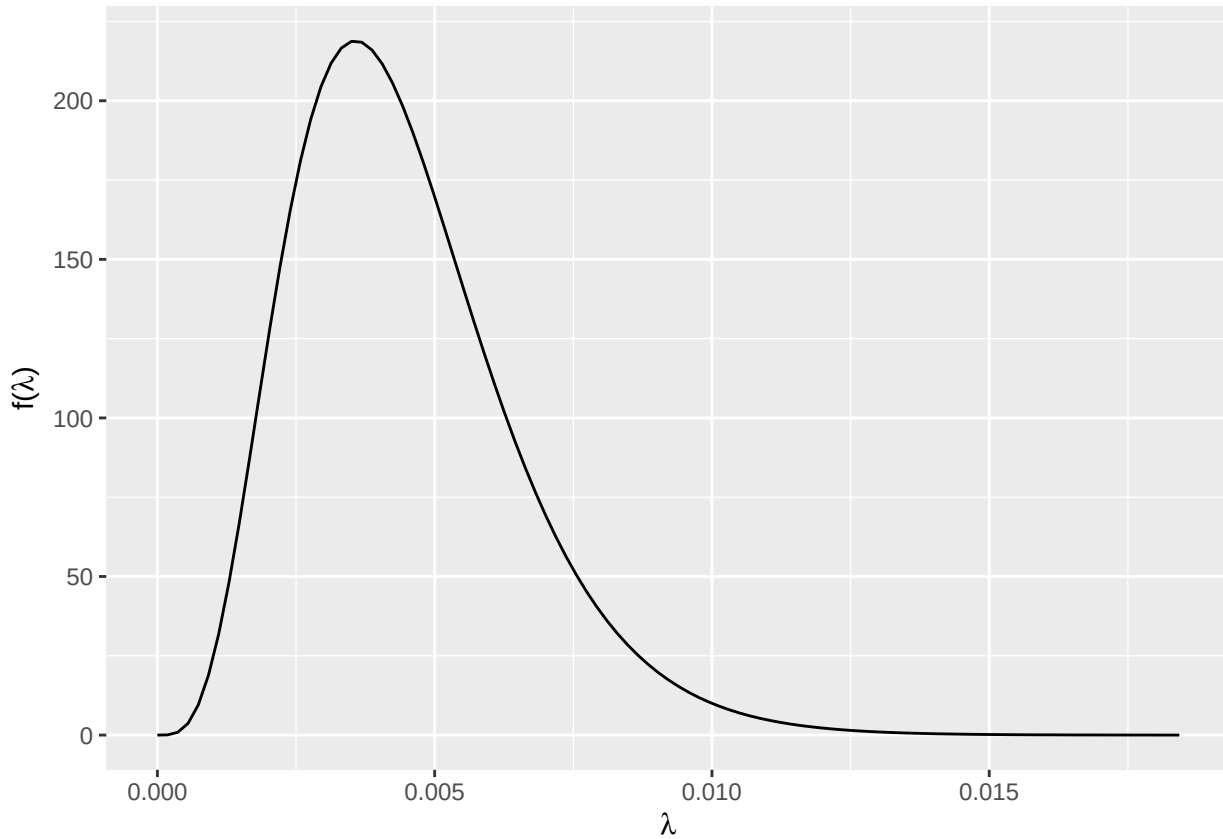
```
## Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(density)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```r
c(shape, rate)
```

```
## [1] 2.000000000 0.004462201
```

```r
theta_exp <- rate
theta_var <- theta_exp^2 / 5
#theta_exp = a/b
#theta_var = a/b^2
b <- theta_exp / theta_var
a <- theta_exp * b

plot_gamma(a, b)
```




```r
data <- read.csv("data/earthquake.csv")
head(data)
```

```
##                       time latitude longitude  depth mag magType nst gap  dmin
## 1 2025-07-29T23:24:52.483Z  52.4948  160.2395  35.00 8.8     mww 251  30 1.102
## 2 2021-08-12T18:35:17.231Z -58.3753  -25.2637  22.79 8.1     mww  NA  78 7.463
## 3 2021-07-29T06:15:49.188Z  55.3635 -157.8876  35.00 8.2     mww  NA  NA    NA
## 4 2021-03-04T19:28:33.178Z -29.7228 -177.2794  28.93 8.1     mww  NA  21 0.739
## 5 2019-05-26T07:41:15.073Z  -5.8119  -75.2697 122.57 8.0     mww  NA  17 3.180
## 6 2018-08-19T00:19:40.670Z -18.1125 -178.1530 600.00 8.2     mww  NA  13 3.630
##    rms net           id                updated
## 1 0.73  us   us6000qw60 2026-02-27T21:52:05.218Z
## 2 0.76  us   us6000f53e 2025-12-22T22:13:38.221Z
## 3 0.95  ak ak0219neiszm 2025-12-23T18:40:37.421Z
## 4 0.69  us   us7000dflf 2025-12-22T18:58:46.851Z
## 5 0.84  us   us60003sc0 2025-12-22T18:09:59.494Z
## 6 0.79  us   us1000gcii 2025-11-26T16:39:52.070Z
##                                          place       type horizontalError
## 1   2025 Kamchatka Peninsula, Russia Earthquake earthquake            6.78
## 2        2021 South Sandwich Islands Earthquake earthquake            8.00
## 3               2021 Chignik, Alaska Earthquake earthquake              NA
## 4 2021 Kermadec Islands, New Zealand Earthquake earthquake            7.70
## 5                     78 km NE of Navarro, Peru earthquake            7.00
## 6                          2018 Fiji Earthquake earthquake            9.60
##   depthError magError magNst   status locationSource magSource
## 1      1.784    0.018    303 reviewed             us        us
```

```
## 2      3.600    0.065     23 reviewed             us        us
## 3      0.000       NA     NA reviewed             ak        ak
## 4      3.300    0.034     81 reviewed             us        us
## 5      4.000    0.036     75 reviewed             us        us
## 6      1.900    0.045     47 reviewed             us        us
```

```
earthquake <- data[, c("time", "mag")]
earthquake$datetime <- as.POSIXct(earthquake$time, format = "%Y-%m-%dT%H:%M:%OSZ", tz = "UTC")
earthquake <- earthquake[order(earthquake$datetime), ]
earthquake$intertime <- c(NA, as.numeric(
  difftime(earthquake$datetime[-1],earthquake$datetime[-nrow(earthquake)], units="days")))
earthquake$Display <- round(earthquake$intertime, 2)
head(earthquake)
```

```
##                        time mag            datetime intertime Display
## 61 1950-08-15T14:09:34.000Z 8.6 1950-08-15 14:09:34        NA      NA
## 60 1950-12-09T21:38:51.980Z 8.2 1950-12-09 21:38:51  116.3120  116.31
## 59 1952-03-04T01:22:49.230Z 8.1 1952-03-04 01:22:49  450.1555  450.16
## 58 1952-11-04T16:58:30.000Z 9.0 1952-11-04 16:58:30  245.6498  245.65
## 57 1957-03-09T14:22:33.000Z 8.6 1957-03-09 14:22:33 1585.8917 1585.89
## 56 1957-12-04T03:37:53.790Z 8.1 1957-12-04 03:37:53  269.5523  269.55
```

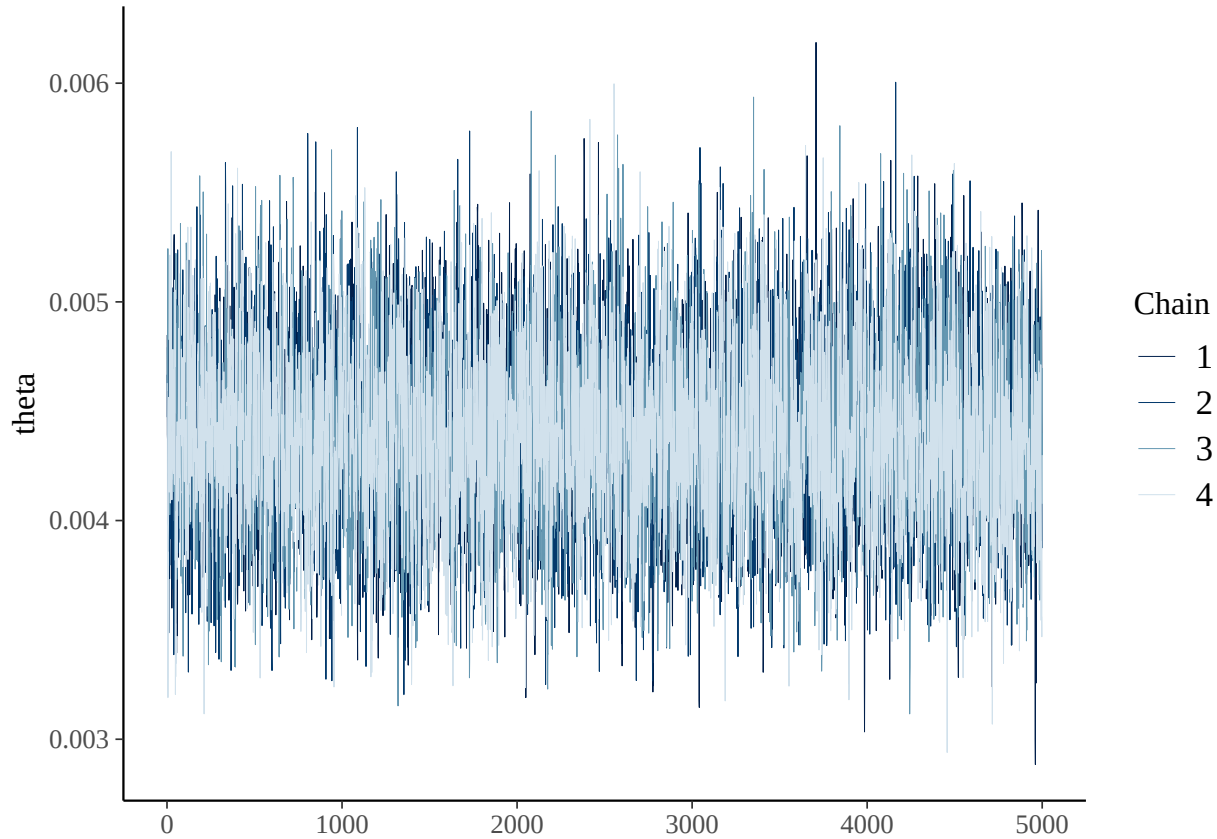
```
tObs <- earthquake$intertime[!is.na(earthquake$intertime)]
n <- length(tObs)
sum <- sum(earthquake$intertime, na.rm = TRUE)
lastDate <- max(earthquake$date)
xDay <- as.numeric(difftime(Sys.Date(), lastDate, units="days"))

#theta ~ Gamma(a,b) prior
c(a, b)
```

```
## [1]    5.000 1120.523
```

```
#posterior
aPost <- a + 2*n
bPost <- b + sum
```

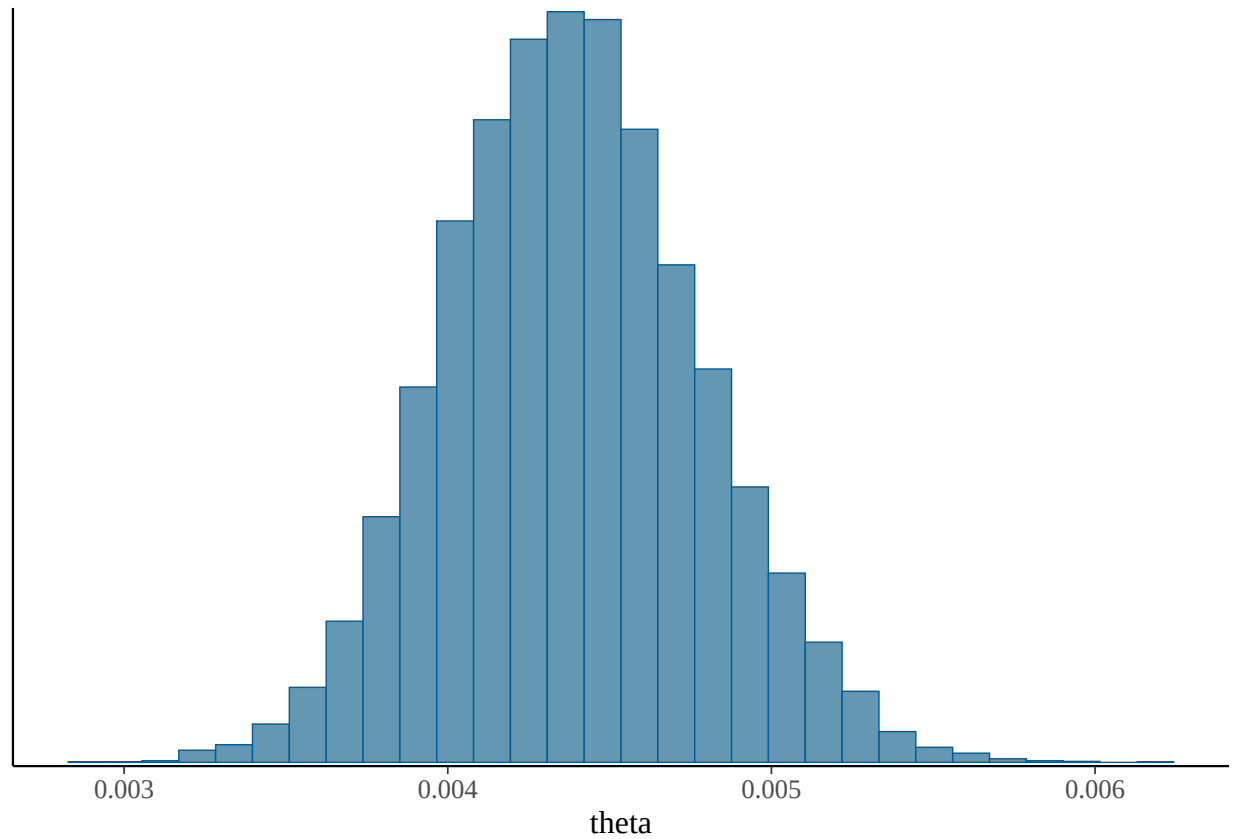
```
model <- "
data {
  int<lower=1> N;
  vector<lower=0>[N] T;
}
parameters {
  real<lower=0> theta;
}
model {
  T ~ gamma(2, theta);
  theta ~ gamma(5.0, 1120.523);
}
"
```

```r
simulation <- stan(model_code=model, data=list(N = n, T = tObs),
                   chains=4, iter=5000*2, seed=1234,
                   refresh=FALSE)
bayesplot::mcmc_trace(simulation, pars="theta", size=0.5)
```

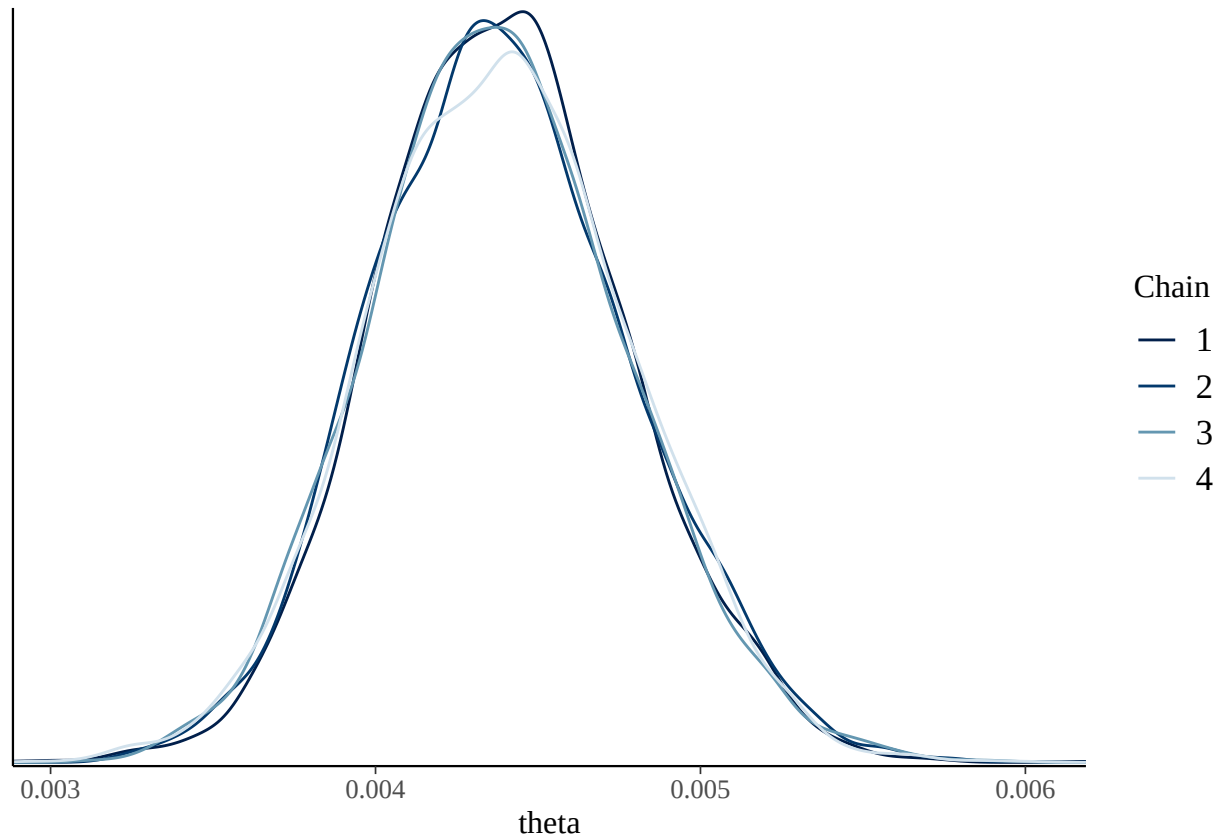



```r
bayesplot::mcmc_hist(simulation, pars="theta")
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
bayesplot::mcmc_dens_overlay(simulation, pars = "theta")
```

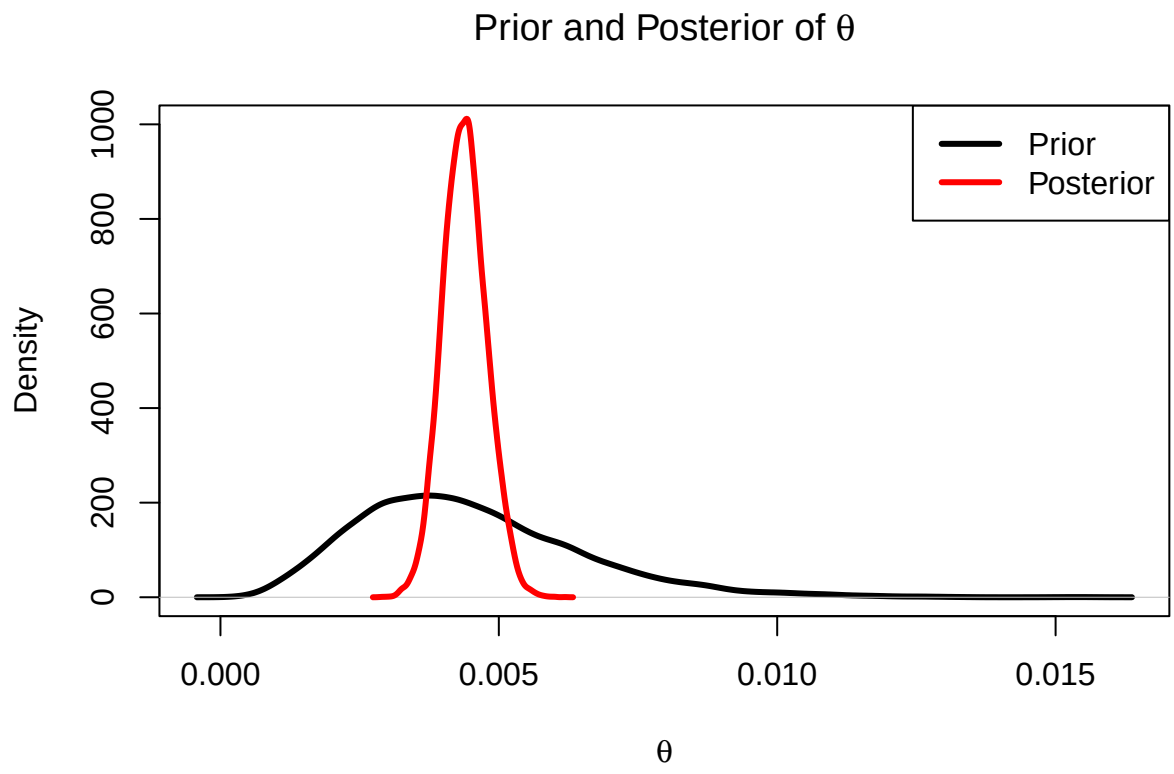



```r
summary(simulation, pars="theta")
```

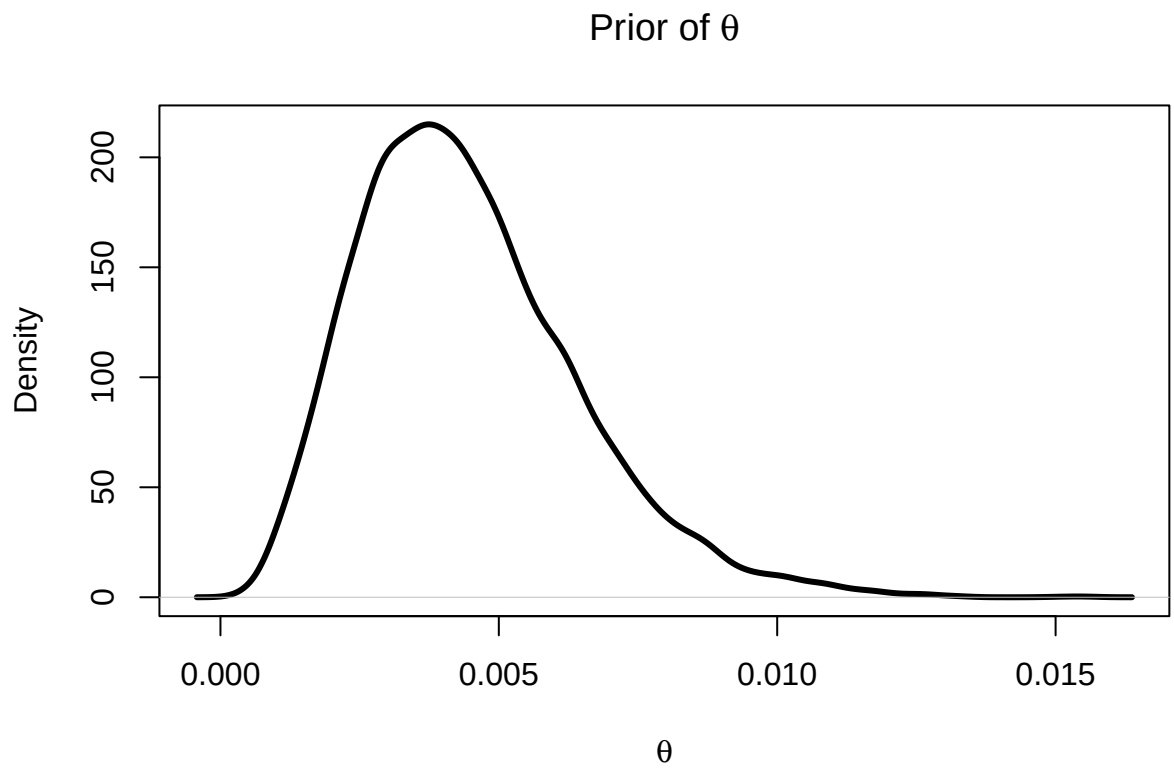
```
## $summary
##              mean      se_mean           sd       2.5%        25%         50%
## theta 0.004386573 4.706145e-06 0.0003933709 0.00364314 0.00411609 0.004378243
##               75%       97.5%    n_eff     Rhat
## theta 0.004643346 0.005181287 6986.729 1.000269
##
## $c_summary
## , , chains = chain:1
##
##          stats
## parameter        mean           sd        2.5%         25%         50%
##     theta 0.004390995 0.0003810079 0.003670363 0.004130059 0.004383876
##          stats
## parameter        75%       97.5%
##     theta 0.00463881 0.005165812
##
## , , chains = chain:2
##
##          stats
## parameter        mean           sd       2.5%         25%         50%
##     theta 0.004390286 0.0003977129 0.00365513 0.004108408 0.004372113
##          stats
## parameter         75%       97.5%
```

```
##     theta 0.004650627 0.005203404
##
## , , chains = chain:3
##
##          stats
## parameter        mean           sd        2.5%         25%         50%
##     theta 0.004378364 0.0003941911 0.003640888 0.004110158 0.004368549
##          stats
## parameter         75%       97.5%
##     theta 0.004633174 0.005187134
##
## , , chains = chain:4
##
##          stats
## parameter        mean           sd        2.5%         25%         50%
##     theta 0.004386646 0.0004002819 0.003615074 0.004110498 0.004387426
##          stats
## parameter         75%      97.5%
##     theta 0.004650516 0.00516943
```

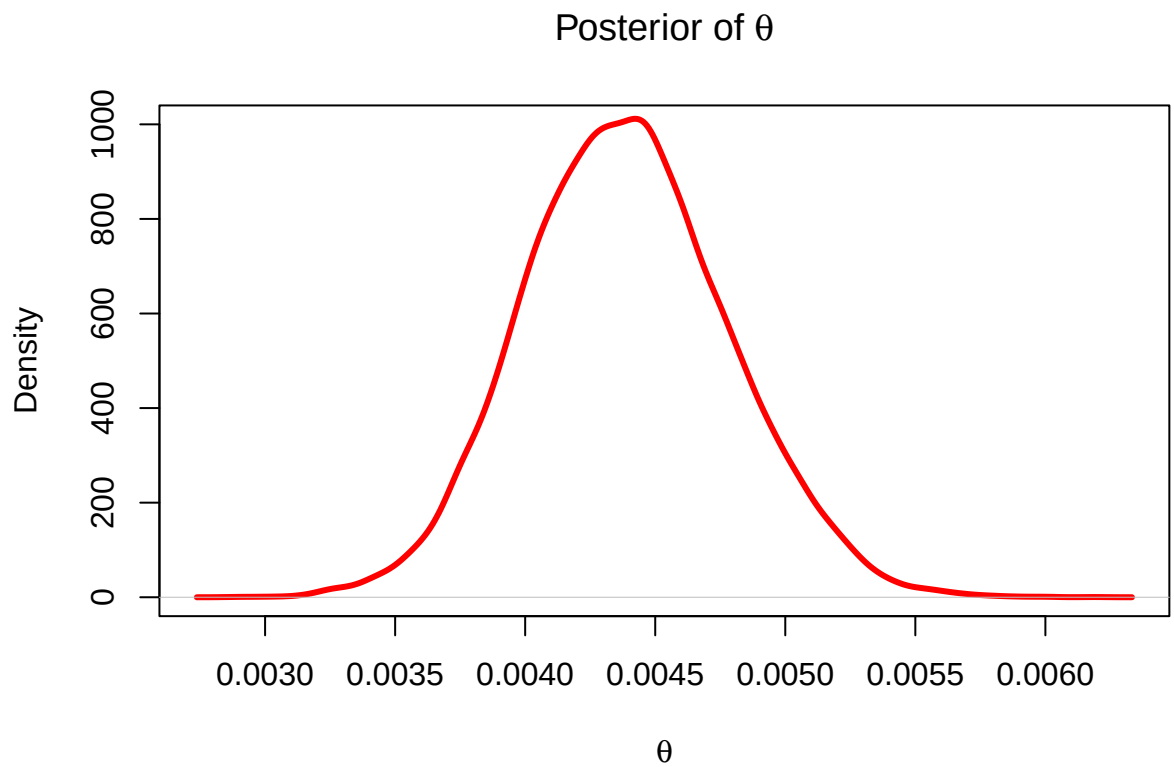
```r
priorTheta <- rgamma(10000, shape = a, rate = b)
postTheta <- as.data.frame(simulation, pars = "theta")$theta
plot(density(priorTheta), lwd = 3, col = "black",
     xlab = expression(theta), ylab = "Density",
     main = expression("Prior and Posterior of " * theta),
     ylim = range(c(0, 1000)))
lines(density(postTheta), lwd = 3, col = "red")
legend("topright", legend = c("Prior", "Posterior"), col = c("black", "red"), lwd = 3)
```




```r
plot(density(priorTheta), lwd = 3, col = "black",
     xlab = expression(theta), ylab = "Density",
     main = expression("Prior of " * theta))
```




```r
plot(density(postTheta), lwd = 3, col = "red",
     xlab = expression(theta), ylab = "Density",
     main = expression("Posterior of " * theta),
     ylim = range(c(0, 1000)))
```



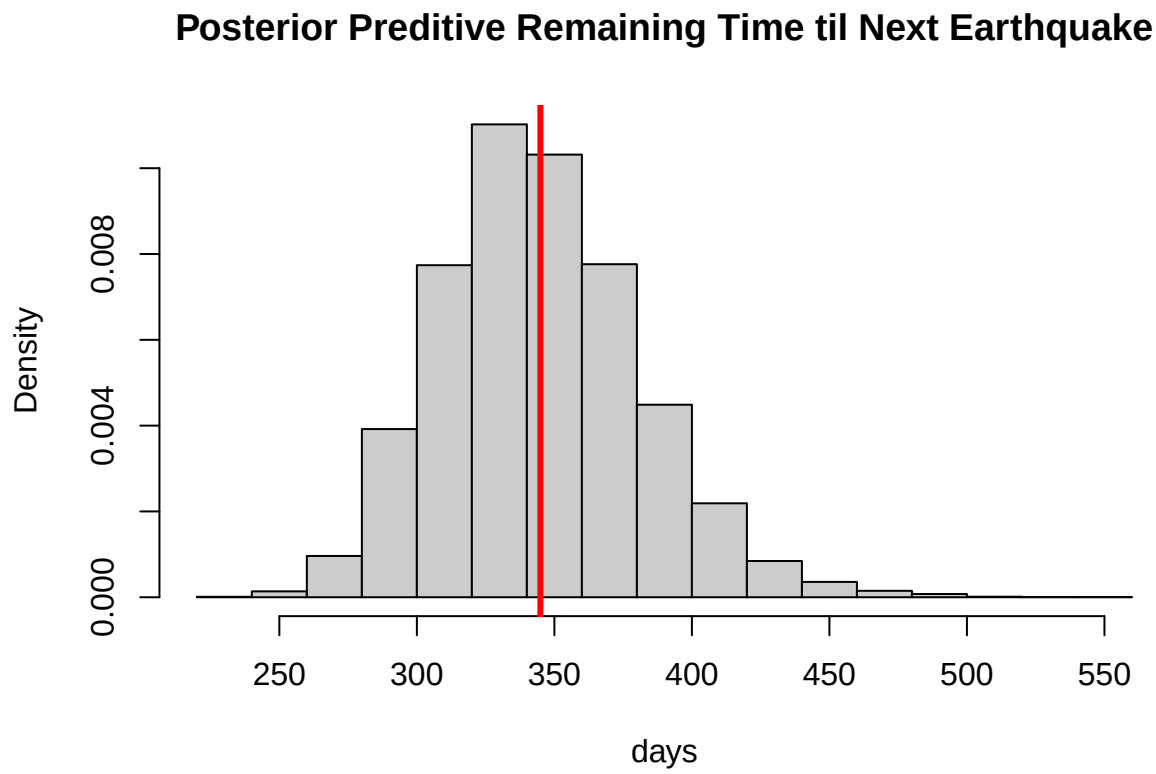

```r
cred_int_95 <- quantile(postTheta, probs = c(0.025, 0.975))
head(cred_int_95)
```

```
##        2.5%       97.5%
## 0.003643140 0.005181287
```

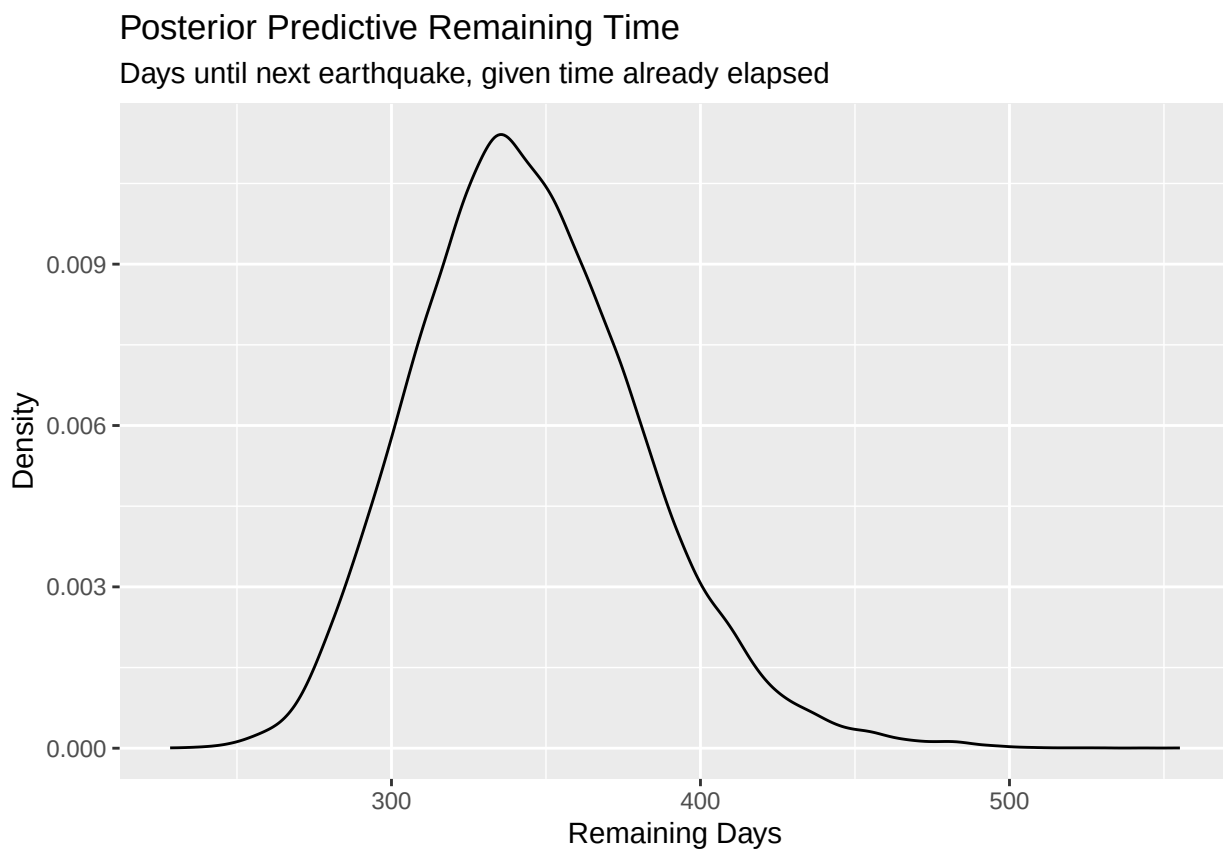
```r
remainDays <- (xDay + 2/postTheta) / (1+postTheta*xDay)
summary(remainDays)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   228.4   319.5   342.2   344.9   367.7   555.2
```

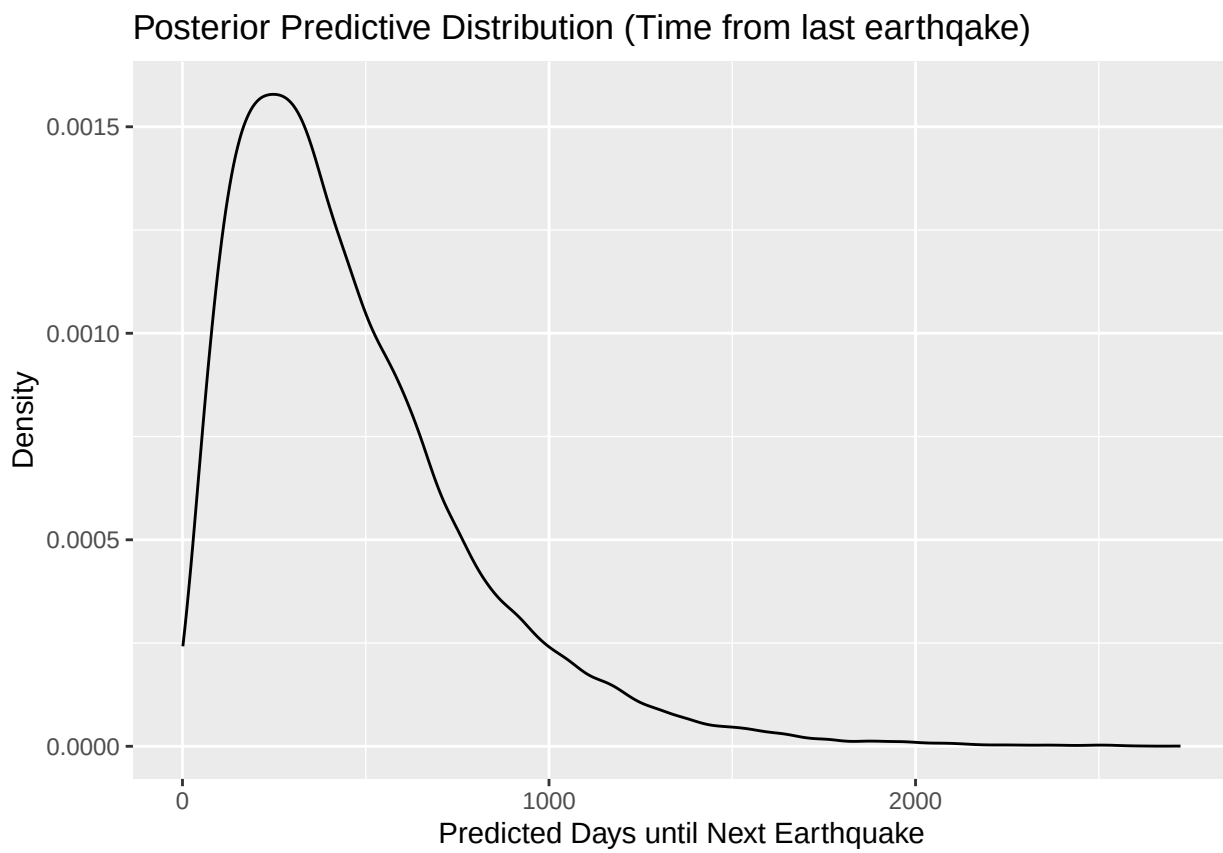
```r
hist(remainDays, prob=TRUE, col="gray80",
     xlab="days",
     main="Posterior Preditive Remaining Time til Next Earthquake")
abline(v=mean(remainDays), col="red", lwd=3)
```



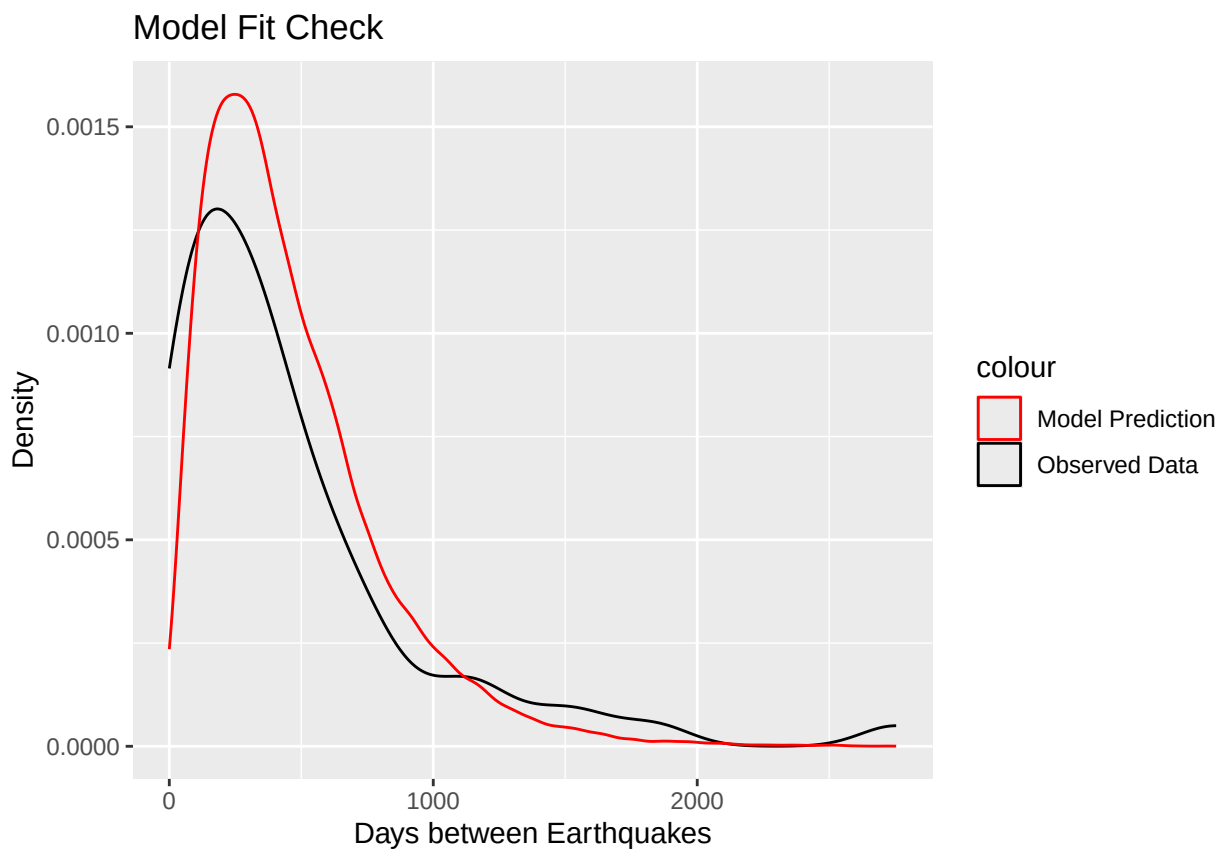

```r
df_remain <- data.frame(days = remainDays)

ggplot(df_remain, aes(x = days)) +
  geom_density() +
  labs(title = "Posterior Predictive Remaining Time",
       subtitle = "Days until next earthquake, given time already elapsed",
       x = "Remaining Days",
       y = "Density")
```




```r
#this is graph is prediction interval relative to last earthquake that happened
check_sim <- as.data.frame(simulation)
check_sim <- check_sim %>%
  mutate(y_predict = rgamma(n = n(), shape = 2, rate = theta))

ggplot(check_sim, aes(x = y_predict)) +
  geom_density() +
  labs(title = "Posterior Predictive Distribution (Time from last earthqake)",
       x = "Predicted Days until Next Earthquake",
       y = "Density")
```



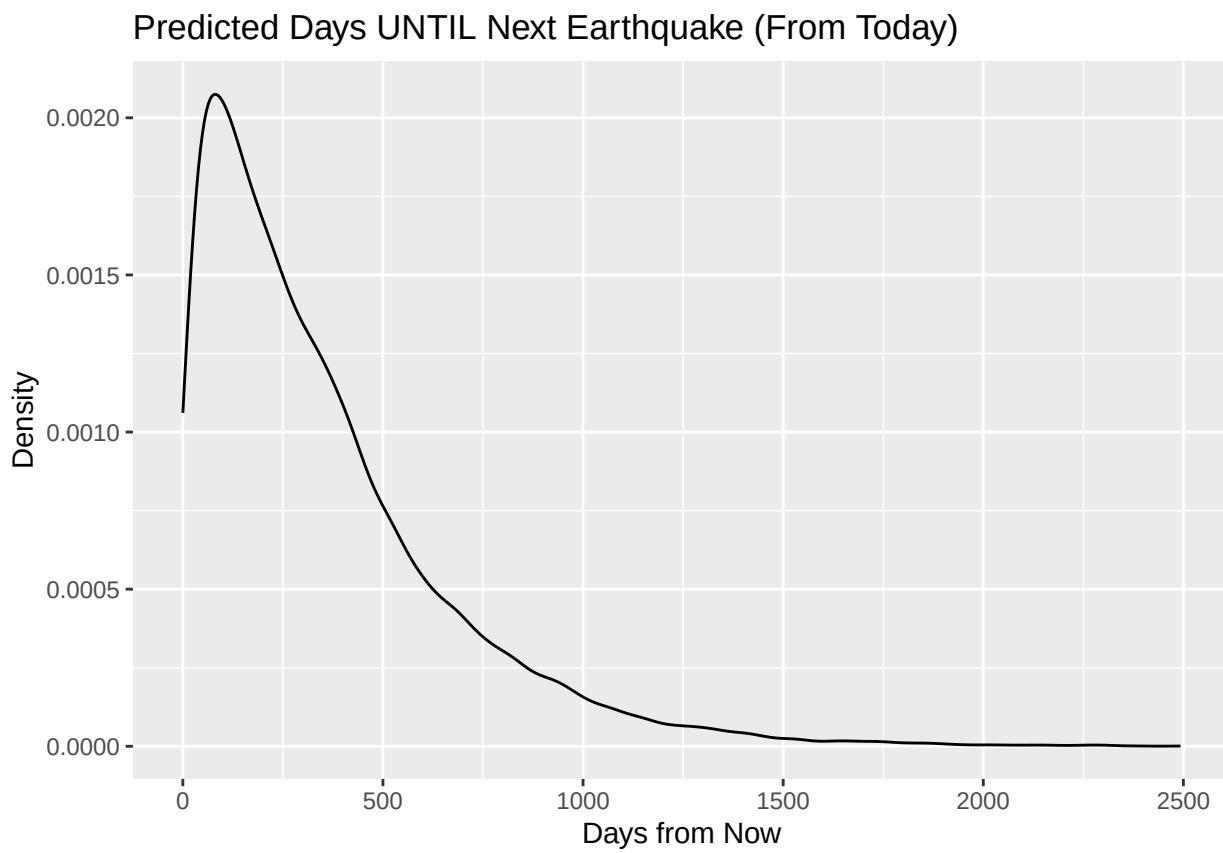

```
prediction_int <- quantile(check_sim$y_predict, probs = c(0.025, 0.975))

ggplot() +
  geom_density(aes(x = tObs, color = "Observed Data")) +
  geom_density(aes(x = check_sim$y_predict, color = "Model Prediction")) +
  labs(title = "Model Fit Check", x = "Days between Earthquakes", y = "Density") +
  scale_color_manual(values = c("Observed Data" = "black", "Model Prediction" = "red"))
```




```r
#this is graph is prediction interval relative today
check_sim <- check_sim %>%
  mutate(days_from_today = y_predict - xDay)

check_sim_future <- check_sim %>%
  filter(days_from_today > 0)

head(check_sim_future)
```

```
##         theta      lp__ y_predict days_from_today
## 1 0.004848658 -804.3083 1077.4519       847.42754
## 2 0.004393150 -803.6592  715.3740       485.34965
## 3 0.004371412 -803.6598  566.3439       336.31953
## 4 0.004785449 -804.1472  452.7497       222.72527
## 5 0.004715919 -803.9953  751.5081       521.48369
## 6 0.004754291 -804.0758  268.7121        38.68771
```

```r
ggplot(check_sim_future, aes(x = days_from_today)) +
  geom_density() +
  labs(title = "Predicted Days UNTIL Next Earthquake (From Today)",
       x = "Days from Now",
       y = "Density")
```




```
prediction_int_today <- quantile(check_sim_future$days_from_today, probs = c(0.025, 0.5, 0.975))
```

```
#should we include this?
```

Given 230.0243926days have passed since the last M8+ earthquake worldwide, the expected remaining waiting time until the next one is about 344.9233025days.

```
daysVec <- quantile(remainDays, probs = c(0, 0.25, 0.5, 0.75, 1))

sapply(daysVec, function(d) {
  probInDays <- (
    pgamma(xDay + d, shape = 2, rate = postTheta) -
    pgamma(xDay, shape = 2, rate = postTheta)
  ) / (
    1 - pgamma(xDay, shape = 2, rate = postTheta)
  )
  mean(probInDays)
})
```

```
##        0%       25%       50%       75%      100%
## 0.4485301 0.5800248 0.6083963 0.6383517 0.8029081
```